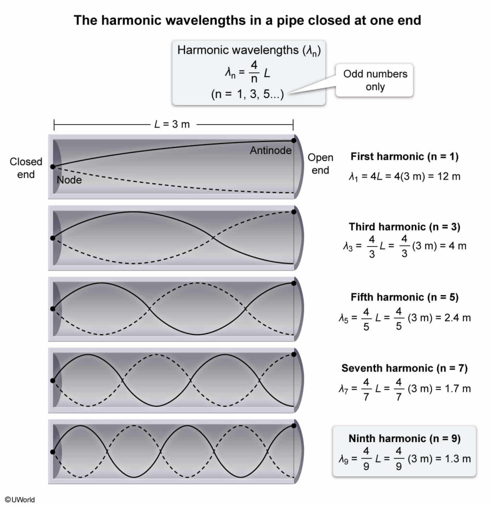
A 3.0 m long organ pipe is closed at one end. Which of the following is a possible wavelength associated with a resonance in the pipe?

- A. 1.3 m [35%]
- B. 3.0 m [36%]
- C. 5.2 m [10%]
- D. 24 m [17%]

Correct
36% answered correctly

Explanation:



Standing sound waves are generated in pipes due to interference between incident waves and reflected waves, which are moving in opposite directions in the air tube. The resonating sound waves in a pipe create nodes (points of zero amplitude) at closed ends due to restricted air movement, and antinodes (points of maximal amplitude) at open ends due to free motion of air molecules.

A pipe with one end open resonates when a node is formed at the closed end, and an antinode is formed at the open end. The fundamental standing wave occurs when the length of the tube L is **one fourth** the wavelength λ of the standing wave:

$$L = \frac{1}{4}\lambda$$

The wavelength of standing waves harmonics, which are generated in a tube closed at one end, have the general formula:

$$\lambda_{\text{n}} = \frac{4}{n}L$$

where n is any odd integer (n = 1, 3, 5, 7…).

In this question, a 3 m pipe with one end closed produces the following first five harmonics:

$$\lambda_1 = 4L = 4(3\text{ m}) = 12\text{ m}$$

$$\lambda_3 = \frac{4}{3}L = \frac{4}{3}(3\text{ m}) = 4\text{ m}$$

$$\lambda_5 = \frac{4}{5}L = \frac{4}{5}(3\text{ m}) = 2.4\text{ m}$$

$$\lambda_7 = \frac{4}{7}L = \frac{4}{7}(3\text{ m}) = 1.7\text{ m}$$

$$\lambda_9 = \frac{4}{9}L = \frac{4}{9}(3\text{ m}) = 1.3\text{ m}$$

Therefore, of the available choices only 1.3 m is a possible wavelength that can produce a standing wave in this pipe.

(Choice B) A standing wavelength that is equal to the pipe's length is not possible because a node must be formed at the closed end and an antinode at the open end.

(Choice C) No standing wavelength values between 12 m and 4 m are possible because the harmonics of a pipe closed at one end occur in odd multiples.

(Choice D) The maximum wavelength for a standing wave in a 3 m pipe closed at one end is 12 m.

Educational objective:
A standing wave in a tube closed at one end has a node at the closed end and an antinode at the open end. The wavelength of standing waves in a tube closed at one end have harmonics with odd multiples of the tube length.

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